MIL-C-46988C (AR)
5 September 1984
SUPERSEDING
MIL-C-46988B (AR)
8 February 1980

MILITARY SPECIFICATION

CARTRIDGE, 105MM, TP-T, M490
LOADING, ASSEMBLING, AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of tank target practice cartridge designated as Cartridge, 105MM, TP-T, M490.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herin.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

8837831 - Box, Packing, Ammunition for Cartridges, 105MM, HEAT and TP-T, for Gun M68, in Fiber Container, M435 or Fiber Container, PA (when directed by the procuring agency).
8837832 - Container, Ammo Fiber, M435 for Cartridges 105MM, HEAT and TP-T for Gun M68 (when directed by the procuring agency)
8865257 - Case, Cartridge, and Liner Assembly, 105MM, M148A1B1
9328579 - Packing and Marking for Container, Ammunition, Fiber, PA 84 for Cartridge 105MM, TP-T, M490
9328580 - Packing and Marking for Box, Pallet Type, Wirebound, for Cartridge 105MM, TP-T, M490
9331331 - Cartridge, 105MM, TP-T, M490

INSPECTION EQUIPMENT DRAWINGS

8828098 - Gage, Profile and Alignment
8828099 - Gage, Alignment Check
8828100 - Gage, Limit Check

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material

3.1.1 General. Materials shall be in accordance with applicable drawings and specifications.

3.2 Cartridges. The cartridges shall comply with all requirements specified on Drawing (dwg.) 9331331 and with all requirements specified in applicable specifications.

3.3 Propellant weight. The weight of the propellant shall not differ by more than plus or minus 0.25 ounce avoirdupois (7.1 grams) from that designated by the loading authorization for the particular lot of propellant used.

3.4 Torque requirements

3.4.1 Assembly torque. Threaded joints shall withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet.

3.4.2 Disassembly torque. Threaded joints shall withstand application of the minimum disassembly torque specified on the applicable drawing, without moving, after thread sealing compound (if required) has dried.

3.5 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.5.6.

3.5.1 Muzzle velocity. The corrected muzzle velocity shall not be greater than 3890 feet per second (1185.7 meters per second) or less than 3810 fps (1161.3 meters per second) and the standard deviation for the lot shall not be greater than 9.5 fps.

3.5.2 Target impact dispersion. The probable error shall not be greater than 0.24 mils horizontal or 0.22 mils vertical.

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3.5.3 Tracer functioning. The tracer shall ignite and there shall be visible evidence of burning of the tracer composition to the 2000 meter target.

3.5.4 Safety. There shall be no hangfire or indication of parts failure in the gun bore or in flight prior to impact during any of the proving ground tests conducted. Obturator band and seal discard is not cause for rejection.

3.6 Process description and controls. The contractor in accordance with the contract provisions shall make available a written description of the specified production process and process controls, determined by the Government as crucial for selected components that are to be used in production of the first article sample (see para. 4.3.4).

3.7 First article inspection. This specification contains technical provisions for a first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship

3.8.1 The cartridge and all components shall be free of cracks, burrs, chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The cartridge shall not be damaged in any manner during the loading operation.

3.8.2 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such paintings and markings have been marred or obliterated. Bare spots in painting, other than marking, are allowable provided that they total not more than 1/4 square inch (1.61 square cm) in aggregate, and that the maximum dimension of any individual bare spot does not exceed 3/16 inch (.48 cm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisons of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article sample

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 55 (see Note 1, Table I) cartridges which have been loaded, assembled and packed using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge, 105MM, TP-T, M490 Assembly				9331331
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for Defects	55 Note 1		3.2	4.4.2.1 thru 4.4.2.11
	Bullet Pull Test	10		3.2	4.5.4
	Ballistic Tests				
	Phase 1 (70°F ± 2.5°F - New Tube - See 6.9)	15			
	Muzzle Velocity			3.5.1	4.5.6.1
	Target-Impact-Dispersion			3.5.2	4.5.6.1
	Tracer Functioning			3.5.3	4.5.6.2
	Safety			3.5.4	4.5.6.3
	Pressure			Info Only	-
	Phase 2 (+125°F - 5°F - New Tube - See 6.9)	15			
	Muzzle Velocity			Info Only	4.5.6.1
	Target-Impact-Dispersion			Info Only	4.5.6.1
	Tracer Functioning			Info Only	-
	Pressure			Info Only	-
	Safety			3.5.4	4.5.6.3

NOTE: 1. After a successful examination of the 55 units is completed, subject 10 samples to the bullet pull test. 15 samples may be used as spotters (see 6.11), but then the total units to be inspected shall be 70. The remaining 45 samples shall be subjected to the ballistic tests.

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Cartridge, 105MM, TP-T, M490 Assembly				9331331
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Phase 3 (-35°F +5°F - Worn Tube - See 6.9)	15			
	Muzzle Velocity			Info Only	4.5.6.1
	Target-Impact-Dispersion			Info Only	4.5.6.1
	Tracer Functioning			Info Only	4.5.6.2
	Pressure			Info Only	-
	Safety			3.5.4	4.5.6.3

NOTES:

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4.3.4 Process controls. At the same time as submission of the First Article Sample, the contractor shall make available a written description of the production process and controls being used in the propellant weighing operation.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with lot provisions of MIL-A-48078. In addition, inspection lots shall contain:

a. Cartridge cases of not more than three lots used from any group of five consecutive lots of one lot interfix from one manufacturer.

b. Projectile metal parts of not more than two consecutive lots of one lot interfix number from one manufacturer.

c. Propellant of one lot from one manufacturer.

d. Loaded primers of not more than two consecutive lots of one lot interfix number from one manufacturer.

e. Loaded tracers of one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Projectile - Prior to Assembly of Tracer Plug & Disc Assembly				10523473
					NEXT HIGHER ASSEMBLY 9331331
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Pettman cement missing from threads of tracer cavity		0.40%	3.2	Visual
102	Tracer missing or reversed		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Projectile - Prior to Assembly of Cartridge Case				10523473
					NEXT HIGHER ASSEMBLY
					9331331
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Tracer plug and disc assembly fails to withstand reapplication of assembly torque		0.40%	3.4	4.5.3
MINOR					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

DRSMC-0A (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Liner, Prior to Assembly Into Cartridge Case				9217050
					NEXT HIGHER ASSEMBLY
					8865257
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR	None defined.				
MINOR					
201	Excessive amount of uncoated surface		0.65%	3.2	Visual
202	Evidence of poor workmanship, including liner film torn or damaged		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Case, Cartridge and Liner Assembly				8865257
					NEXT HIGHER ASSEMBLY 9331331
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Liner installed incorrectly (wrong surface is exposed)		0.40%	3.2	Visual
102	Adhesive missing		0.40%	3.2	Visual
103	Gap between liner and case, excessive		100%	3.2	Visual
104	Depth of liner from mouth of case		0.40%	3.2	Gage
105	Evidence of contamination of cartridge case prior to liner insertion		100%	3.2	Visual
MINOR					
201	Ends of cloth overlap		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Primer, Prior to Assembling Into Cartridge Case and Liner Assembly				8847476
					NEXT HIGHER ASSEMBLY
					9331331
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	One or more flash holes missing		100%	3.2	Visual
MAJOR					
101	Gasket O-ring missing from primer		0.40%	3.2	Visual
102	Deterioration of primer (if applicable)		Note 1	3.2	4.4.3.1
MINOR					
201	None defined.				

NOTES:
1. See applicable primer specification.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Cartridge Case and Liner Assembly, After Assembly of Primer, but Prior to Loading of Propellant		9331331
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Primer above flush		100%	3.2	Straight-Edge
102	Propellant weight	4	4.4.3.2	3.3	Scale
103	Primer fails to withstand minimum disassembly torque		0.40%	3.4	4.5.3
104	Height of primer electrode from front face of case flange, min.		0.40%	3.2	Gage
105	Adhesive missing from mouth of case (after loading of propellant)		0.40%	3.2	Visual
106	Foreign matter in case		0.40%	3.2	Visual
107	Electrical resistance between primer electrode and cartridge case		100%	3.2	4.5.2.1
108	Primer stakes missing or improper		100%	3.2	Visual
109	Primer bent or damaged		100%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Cartridge Assembly				9331331
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Safety	Note 1		3.5.4	4.5.6.3
MAJOR					
101	Profile and alignment		100%	3.2	4.5.5
102	Bullet pull	10	Note 2	3.2	4.5.4
103	Muzzle velocity, target-impact-dispersion, tracer functioning	15	4.4.3.4.1 4.4.3.4.2 4.4.3.4.3	3.5.1 3.5.3 3.5.2	4.5.6.1 & 4.5.6.2
MINOR					
201	Marking missing, incorrect or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: 1. Rounds fired for muzzle velocity shall be observed for compliance with safety requirements.
2. See paragraph 4.4.3.3.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Unsealed Fiber Container (See 5.3)				8837832 or 9328528
					NEXT HIGHER ASSEMBLY
					8837831 or 9328579
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Packing component missing or out of place		0.40%	3.2	Visual
MINOR					
201	container cap cannot be removed by hand		0.65%	3.2	Manual
202	Cartridge cannot be removed by hand		0.65%	3.2	Manual
203	Glue or asphalt on cartridge		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Sealed Fiber Container (See 5.3)				8837832 or 9328579
					NEXT HIGHER ASSEMBLY 8837831 or 9328580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Container cut or damaged through all impregnated layers		0.40%	3.2	Visual
102	Metal end loose or distorted		0.40%	3.2	Visual/Manual
103	Tape incomplete or badly wrinkled		0.40%	3.2	Visual
MINOR					
201	Tear tab length, min.		0.65%	3.2	Gage
202	Contents loose		0.65%	3.2	Manual
203	Gap between cover and body of container exeeds 1/8 inch		0.65%	3.2	Visual
204	Cuts, scuffs, or gouges in outer layers		0.65%	3.2	Visual
205	Marking misleading or identifiable		0.65%	3.2	Visual

NOTES:

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MIL-C-46988C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Sealed Wood Packing Box (See 5.3)				8837831
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Box damaged exposing contents		0.40%	3.2	Visual
102	Board(s) loose or broken		0.40%	3.2	Visual
103	Hardware or strapping missing, broken or loose		0.40%	3.2	Visual/Manual
104	DoD code or other marking missing, incorrect or unidentifiable		0.40%	3.2	Visual
MINOR					
201	Handle missing or insecure		0.65%	3.2	Visual/Manual
202	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Box, Pallet Type, Wirebound				9328580
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Box damaged		0.40%	3.2	Visual
102	Boards loose or broken		0.40%	3.2	Visual
103	Hardware or strapping missing, broken or loose		0.40%	3.2	Visual/Manual
104	DoD code or other marking missing, incorrect or unidentifiable		0.40%	3.2	Visual
MINOR					
201	Seal missing, unsealed or improperly positioned		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Check test for possible deterioration of primer. See applicable primer specification. If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test specified in the applicable primer loading specification immediately before the primer lot is assembied into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government Inspector at the facility assembling the cartridge (see 6.7).

4.4.3.2 Propellant weight. The propellant shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing using a different balance from that used to make the orginal weighing, and if performed manually, shall be performed by another operator. Any propellant charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.4.4.

4.4.3.3 Bullet pull. The sampling plan for this test shall be in accordance with Standard MIL-STD-414. Section B, Part II, Table B-3, Code Letter F and an AQL of 1.50 percent shall be used. The test shall be performed as specified in 4.5.4 with equipment in accordance with 4.4.4.

4.4.3.4 Proving ground. The following tests shall be performed as specified in 4.5.6. These tests shall be conducted at a Government proving ground in accordance with the applicable approved Acceptance Test Procedure. The sample cartridges shall be fired from a tank mounted M68 gun and observation made for conformance to all applicable requirements.

4.4.3.4.1 Muzzle velocity. The sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code Letter G, and an AQL of 1.0%. In addition, if the standard deviation of the sample cartridges multiplied by the factor 0.70 exceeds the lot requirement, the lot shall be rejected.

4.4.3.4.2 Tracer functioning. The sample cartridges fired in the test for muzzle velocity (see 4.4.3.4.1) shall be observed for tracer functioning. If two or more sample cartridges fail to comply with the applicable requirements, the lot shall be rejected.

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4.4.3.4.3 Target-impact-dispersion. The sample cartridges fired in the test for muzzle velocity (see 4.4.3.4.1) shall be observed for target-impact-dispersion as specified in 4.5.6.1. If the calculated probable error (see 6.10) of the test samples fired multiplied by the appropriate factor (see 6.10.3) below exceeds the lot requirement, the lot shall be rejected.

Sample Size	Degrees of Freedom	Factor
15 - Test Rds.	12	.68
14 (one target miss)	11	.67
10 - Calibration Rds.	8	.64
9 (one target miss)	7	.62

4.4.3.4.4 Safety. All cartridges fired in the proving ground tests shall be observed for hangfires or break-up of parts. If any cartridge fails to comply with the applicable requirement the lot shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraphg Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

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4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Electrical tests

4.5.2.1 Electrical resistance of primer circuit. The resistance shall be measured on the cartridge case-primer assembly, prior to loading of propellant, using an electrical circuit which limits the current through the primer to 20 milliamperes, max., with the primer electrode as one contact and the cartridge case as the other. Any assembly that fails to meet the requirement shall be cause for rejection and removed from the lot.

4.5.2.2 Electrical test equipment. All electrical test equipment required to perform the tests in 4.5.2 shall be checked prior to each shift. This shall be accomplished using actual components or a device that simulates acceptable or rejectable components.

4.5.3 Torque testing. The torque shall be applied in a tightening direction for assembly torques, and in a loosening direction for disassembly torques, while holding the body for projectile torque tests, or the cartridge case for the primer torque test. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.5.4 Bullet pull. The sample cartridges shall be tested for security of crimp of the cartridge case by applying the load in an axial direction. The rate of travel of the testing machine head shall be .100 to .125 inch (.25 to .32 cm) per minute, inclusive.

4.5.5 Profile and alignment check. The profile and alignment of each crimped cartridge shall be verified by a gage corresponding to dwg. 8828098, which is to be horizontal (0° + 2.5°). The cartridge shall be inserted with a peak force not to exceed 60 pounds (27.2 Kg) and a peak velocity of 2 ft/sec (61 cm/sec). Any cartridge which fails to be inserted or seated with the required force and velocity shall be rejected. The profile and alignment gage shall not be utilized as an alignment tool.

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4.5.6 Proving ground

4.5.6.1 Muzzle velocity and target-impact-dispersion phase

4.5.6.1.1 Firing sequence. The data generated in this phase will primarily be used for determination of acceptance from a target impact dispersion and muzzle velocity standpoint, and will require the firing of calibration rounds to establish a velocity correction. The firing sequence will be five calibration rounds, fifteen test rounds, and five calibration rounds. This phase will be fired using a tank mounted M68 new gun tube, unless otherwise specified in Table I (see 6.9). A fixed mount may be used providing that the procedure of 4.5.6.1.5 or its equivalent, is able to remove the backlash from the fire control system.

4.5.6.1.2 Target. Target distance is to be 2000 meters and target size is to be 20 feet by 20 feet (6 meters by 6 meters) minimum.

4.5.6.1.3 Temperature conditioning. The ammunition fired in this phase shall be preconditioned to the temperature of 70° ± 2.5°F (21° ± 1.5°C). However, during the first article testing the ammunition fired shall be preconditioned to the temperatures specified in Table I. All propellant shall be stabilized at a uniform temperature prior to firing. This normally requires a period of 24 hours.

4.5.6.1.4 Weapon zeroing - boresight and superelevation. Actual superelevation of the weapon shall be measured and recorded for all target dispersion test phases. This requires that the weapon be zeroed at the beginning, end and during any extended breaks (30 minutes or longer) in target impact test phases using a conventional collimated M26 Pye Watson boresight (or equivalent). Record the weapon elevation and azimuth which aligns the boresight zero on the target center. Then record weapon elevation and azimuth data for all test and calibration phases. The gun superelevation is the weapon elevation during the test phases minus the weapon elevation which produced a muzzle boresight on target center. This data shall be obtained when using either a fixed mount or a service tank weapon system. This information shall be recorded for information only and included in the ballistic test report.

4.5.6.1.5 Gun tube backlash removal. After loading and locking a round in the chamber, the gun tube shall be traversed off the target center, dropped down below the target center, traversed to the opposite side of the target center, elevated to

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the target center elevation, and traversed onto the target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above or beyond the target center, do not reverse direction of gun tube movement, but repeat the sequence.

4.5.6.1.6 Spotting the target. Spotter rounds (see 6.11) will be used to assure the weapon is zeroed to assure engagement of the target center. A minimum of three consecutive target hits within a circle of a one mil radius from the target center by spotters fired at the same elevation and deflection are required. Once the gun has been spotted on the target for a shot group, the elevation and deflection of the weapon shall remain the same for that shot group.

4.5.6.1.7 Firing groups. The test samples shall be fired in five round groups (or occasions), with a cadence of no less than three minutes between rounds and not more than five minutes between rounds. If more than a thirty minute lapse occurs between group firings or if there was an indication of a trend in the previous group's target impacts to move away from the center of the target, the weapon shall be re-zeroed on target center. A cadence of less than three minutes will be allowed, providing it can be maintained until all test and calibration rounds are fired.

4.5.6.1.8 Target impact measurements. Target impacts (including spotters) shall be measured to the nearest one-half inch (centimeter) in both the horizontal and vertical distance from target center. Vertical and horizontal hit pattern results shall be expressed as probable errors in mils and shall be recorded on the firing record. No more than one lot acceptance test shall be performed on each target.

4.5.6.1.9 Wind velocity. Wind velocity and gusts shall be measured at the firing site prior to the firing of each round. The velocity/target-impact-dispersion portion of this ballistic test shall not be fired if surface wind (nominal) exceeds 10 knots (18.5 km/hr.) and surface wind gusts exceeds 5 knots (9.25 km/hr.). In addition, peak surface wind velocity (nominal plus gusts) is not to exceed 10 knots (18.5 km/hr.) during the firing of any shot group during the velocity/target-impact-dispersion phase. Wind data is not required for temperature extreme test samples.

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4.5.6.1.10 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, fifteen additional cartridges, as required, shall be tested. Results of retests shall be referred to the contracting officer for disposition.

4.5.6.1.10.1 Target-impact-dispersion retest. A lot shall be retested whenever the lot fails to meet the target-impact-dispersion requirements in the presence of one or more of the following conditions:

a. The wind velocity exceeds 10 knots (18.5 km/hr) during firing of any test group.

b. The cadence between any two consecutive rounds in a test group exceeds 5 minutes.

c. A change was made in the azimuth or elevation of the weapon within a test group.

d. Both the test samples and the calibration samples fail to meet the target-impact-dispersion requirements.

e. If one of the test samples missed the target, the lot shall be retested if the pooled P.E. of the remaining samples multiplied by the appropriate factor (see 6.10.3) meets the requirements. Two or more target misses by the test samples shall reject the lot. Target miss coordinates shall be estimated and recorded along with pertinent flight observations.

4.5.6.1.10.2 Velocity retest. A lot shall be retested whenever the lot fails to meet the velocity requirements in the presence of one or more of the following conditions:

a. The muzzle velocity correction factor computed from the average muzzle velocity of the calibration samples exceeds 25 fps (7.62 mps).

b. Two or more velocities are lost.

c. The wind velocity exceeds 10 knots (18.5 km/hr) during the firing.

d. Both the test samples and calibration samples fail to meet the velocity requirements.

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4.5.6.2 Tracer functioning. Cartridges fired during the velocity and target-impact-dispersion phase shall be observed for tracer functioning to the 2000 meter target. The distance over which the tracer burns shall be recorded to the nearest estimated 500 meters (i.e., failure, 500 meters, 1000 meters).

4.5.6.3 Safety. All rounds fired in the proving ground test shall be observed for hangfires and other apparent safety hazards.

5. PACKAGING

5.1 Preservation and packaging

5.1.1 Level A. The cartridge shall be preserved and packaged in accordance with dwg. 9328579 (see 5.3).

5.2 Packing

5.2.1 Level A. The cartridges in containers shall be packed in accordance with dwg. 9328580 (see 5.3).

5.3 Alternative packaging and packing. The cartridges may be packaged and packed in accordance with drawings 8837832 and 8837831 when directed by the procuring agency.

6. NOTES

6.1 Intended use. The cartridge covered by this specification is intended for use with tanks with the M68 gun tube.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAT-I(D), Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armamement Research and Development Center (US Army ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Tehnical data originally prepared by these activities is now under the cognizance of US Army ARDC.

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6.5 Submission of test data and ammunition data cards. In addition to the normal distribution of records when the cartridge is procured by the Army Armament, Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, US Army ARDC, ATTN: DRSMC-LCU-CV(D) and DRSMC-QAT-A(D), Dover, NJ 07801.

6.6 Size of inspection lots. It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items for delivery may be equivalent to such contract quantities provided the lot formation criteria and sampling provisions of this specification are maintained.

6.7 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers, or to the contractor assembling the cartridge, and shall not consititute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.8 Firing record information

6.8.1 Target impact plots and meteorological data shall be included in all firing records.

6.8.2 The firing records shall include the serial numbers of all weapon and fire control components, velocity and weapon to target azimuth. Stock component lots of warmers and spotters shall be recorded as well as the lot numbers of reference components.

6.8.3 Each round fired during the test shall be identified in the sequence of firing. For each round fired in the muzzle velocity/target-impact-dispersion phase, the record shall contain deflection, elevation, time of firing, muzzle velocity, tracer functioning, horizontal and vertical coordinates of impact, and flight characteristics. Chamber pressure shall be recorded for all reference rounds fired and all rounds fired during first article testing.

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6.9 Gun tube wear limits. For the purpose of this test a new tube is defined as having no greater than .015 inches primary land wear and a midlife tube as having no greater than .030 inches primary land wear. A worn tube is defined as follows:

Primary Land Wear Limits	Secondary Land Wear Limits	Secondary Groove Wear Limits
Greater than 0.030 in. and less than .056 in.	Between 70% and 140% of primary land wear.	Between 80% and 190% of primary land wear.

6.10 Method of calculating pooled PE. The individual PE results of the groups (5 samples in each group) of samples shall be pooled in accordance with the following formula:

$$PE_{pooled} = \sqrt{\frac{\sum \left[(N_i - 1) \quad PE_i^2 \right]}{\sum (N_i - 1)}}$$

Where: i = Number of groups
N_i = Number of samples in a group (target hits)
PE_i = Probable Error of the group

6.10.1. Probable error calculations. The probable error of the group is calculated in accordance with the following equation:

$$PE_{mils} = \frac{(.6745)\ (1000)\ \ (SD)}{(X)\ \ (Rg)}$$

Where:
SD - Standard deviation
Rg - Range in meters
X - Conversion factor
(a) when target hits measure in inches, X = 39.37
(b) when target hits are measured in centimeters, X = 100

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6.10.2 Standard deviation. The standard deviation of a sample is calculated in accordance with the following equation found on page 1-10 AMCP 706-110:

$$SD = \sqrt{\frac{N\Sigma X^2 - (\Sigma X)^2}{N\ (N-1)}}$$

Where:
N - the number in the sample
X - the individual data point

6.10.3 Probable error factor. The factors listed in 4.4.3.4.3 are taken from Table A-21 of AMC Pamphlet 706-114 using the 99 percent upper confidence limit (a .01) and (N_i - K) degrees of freedom.

6.11 Spotters. The spotter rounds used during any ballistic test shall ballistically match the test rounds. The weight of the assembled spotter round shall match the weight of the round to be tested. Five spotter rounds shall be available at the proving ground during each production lot acceptance test. Fifteen spotters shall be available at the proving ground during the first article ballistic test. Spotter rounds used shall be of the same configuration as the test rounds.

6.12 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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